A1.2 Nucleic acids

A1 Unity and diversity
Molecules

SL

Understandings:

- A1.2.1—DNA as the genetic material of all living organisms
- A1.2.2—Components of a nucleotide
- A1.2.3—Sugar–phosphate bonding and the sugar–phosphate “backbone” of DNA and RNA
- A1.2.4—Bases in each nucleic acid that form the basis of a code
- A1.2.5—RNA as a polymer formed by condensation of nucleotide monomers
- A1.2.6—DNA as a double helix made of two antiparallel strands of nucleotides with two strands linked by hydrogen bonding between complementary base pairs
- A1.2.7—Differences between DNA and RNA
- A1.2.8—Role of complementary base pairing in allowing genetic information to be replicated and expressed
- A1.2.9—Diversity of possible DNA base sequences and the limitless capacity of DNA for storing information
- A1.2.10—Conservation of the genetic code across all life forms as evidence of universal common ancestry

Key words:

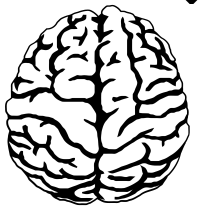
Deoxyribonucleic acid
Nucleotides
Phosphate group
Nitrogenous base
Pentose sugar
Ribose
Deoxyribose
Phosphate-sugar backbone
Adenine
Cytosine
Guanine
Thymine
Uracil
Codon
Condensation
Double helix
Antiparallel
Replication
Transcription
Universal

IB Learner Profile

Inquirers - We nurture our curiosity, developing skills for inquiry and research. We know how to learn independently and with others. We learn with enthusiasm and sustain our love of learning throughout life.



Knowledgeable - We develop and use conceptual understanding, exploring knowledge across a range of disciplines. We engage with issues and ideas that have local and global significance.



Thinkers - We use critical and creative thinking skills to analyse and take responsible action on complex problems. We exercise initiative in making reasoned, ethical decisions



Communicators - We express ourselves confidently and creatively in more than one language and in many ways. We collaborate effectively, listening carefully to the perspectives of other individuals and groups.



Principled - We act with integrity and honesty, with a strong sense of fairness and justice, and with respect for the dignity and rights of people everywhere. We take responsibility for our actions and their consequences.



Open minded - We critically appreciate our own cultures and personal histories, as well as the values and traditions of others. We seek and evaluate a range of points of view, and we are willing to grow from the experience.



Caring - We show empathy, compassion and respect. We have a commitment to service, and we act to make a positive difference in the lives of others and in the world around us.



Risk takers - We approach uncertainty with forethought and determination; we work independently and cooperatively to explore new ideas and innovative strategies. We are resourceful and resilient in the face of challenges and change.



Balanced - We understand the importance of balancing different aspects of our lives—intellectual, physical, and emotional—to achieve well-being for ourselves and others. We recognize our interdependence with other people and with the world in which we live.



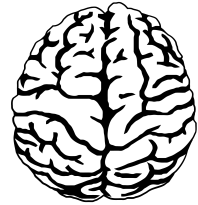
Reflective - We thoughtfully consider the world and our own ideas and experience. We work to understand our strengths and weaknesses in order to support our learning and personal development.



Guiding questions

How does the structure of nucleic acids allow hereditary information to be stored?

How does the structure of DNA facilitate accurate replication?



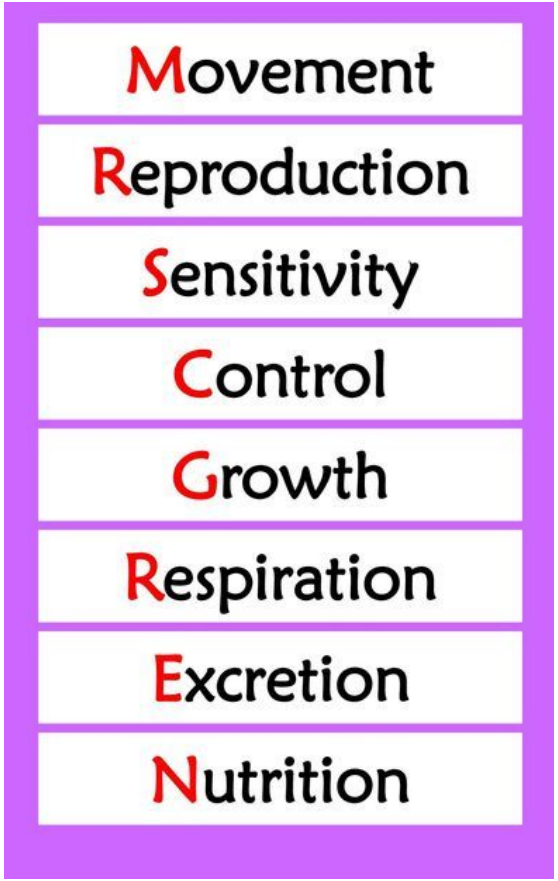
What are the characteristics of all living things?

What are the characteristics of all living things?

What about being made of cells?

Is there any living thing that is not made of cells?

What do all living cells (at some point in their existence point) contain?



A1.2.1—DNA as the genetic material of all living organisms

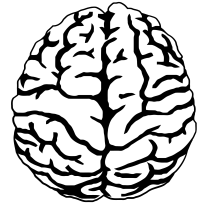


Deoxyribonucleic acid is an essential component of all living things.

All life arises from the instructions present in DNA.

Some viruses contain DNA, but some contain RNA.

Are viruses living things?



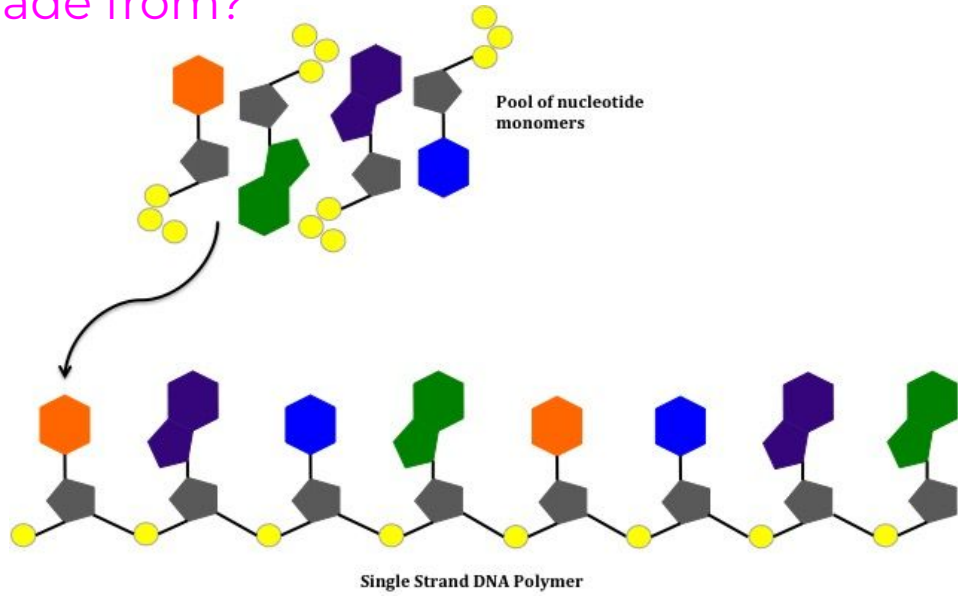
DNA is a polymer (a chain of monomers linked together)

What is the monomer that DNA is made from?

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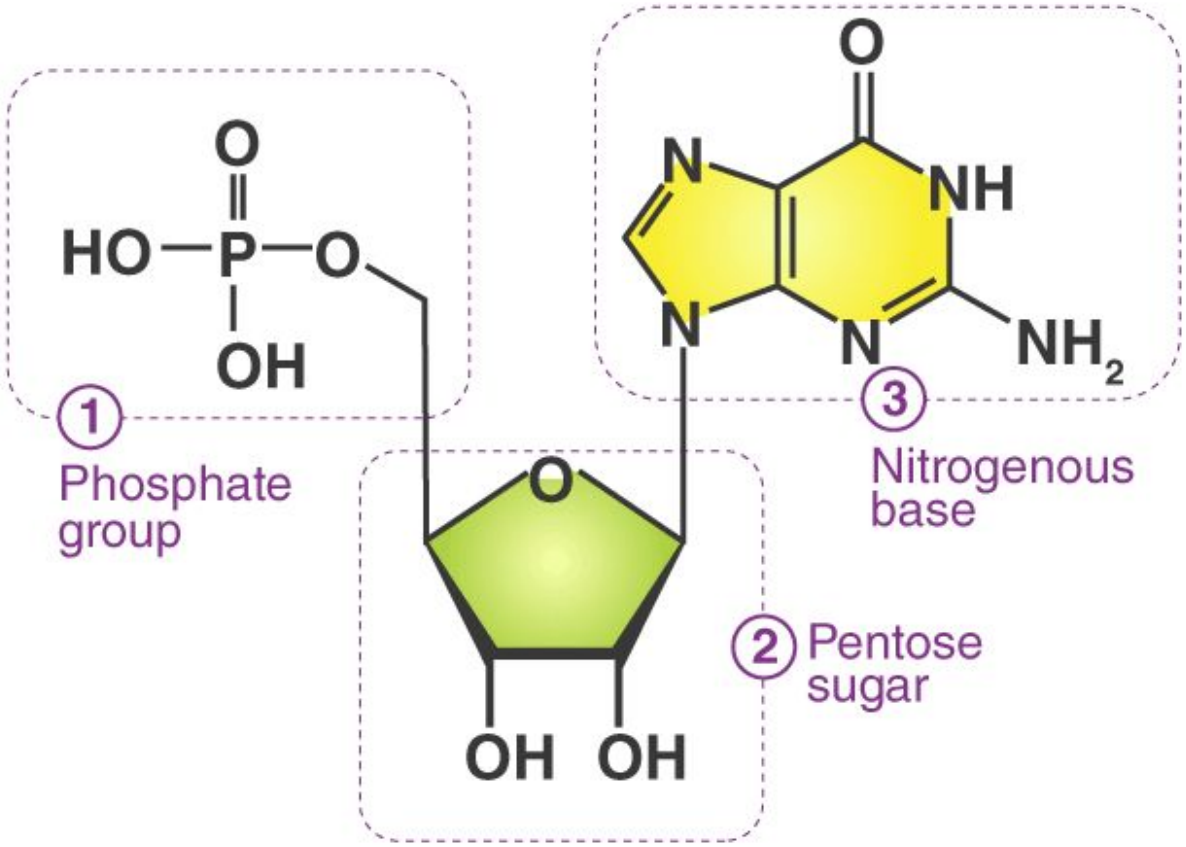
What is the monomer that DNA is made from?

Nucleotides!

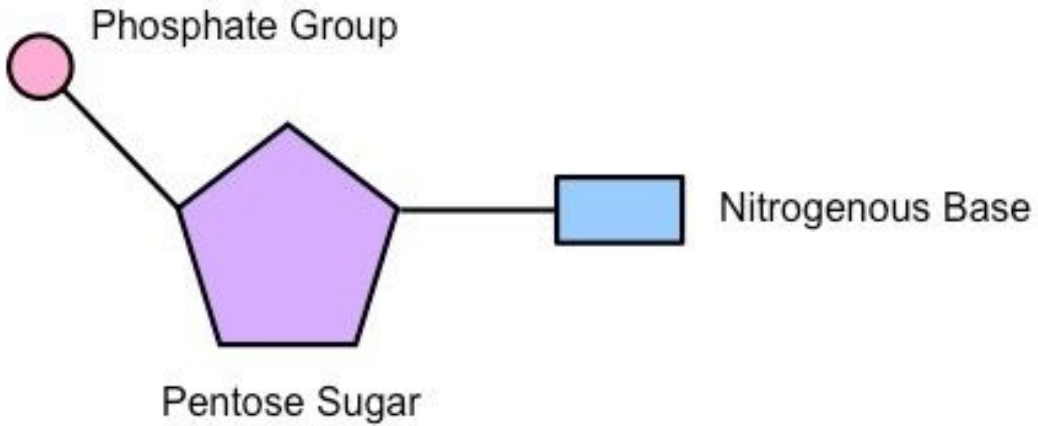


A1.2.2—Components of a nucleotide

NUCLEOTIDE

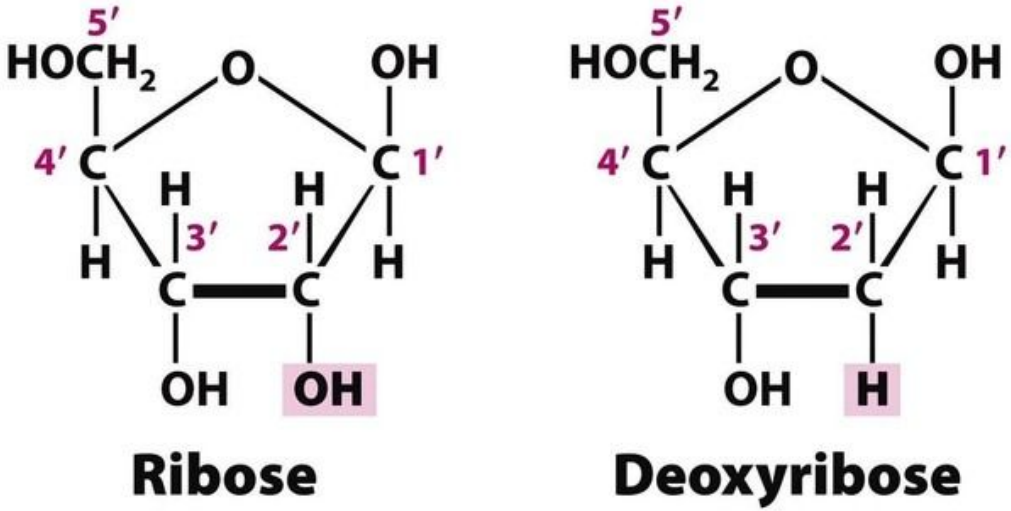

O
HO
P
O
OH
1
Phosphate group
O
N
NH
N
N
NH2
3
Nitrogenous base
O
OH
OH
2
Pentose sugar

How to draw a nucleotide in IB:



The pentose sugar in a nucleotide can be ribose (in RNA) or deoxyribose (DNA)

The carbons are labelled 1-5 (clockwise from the oxygen.



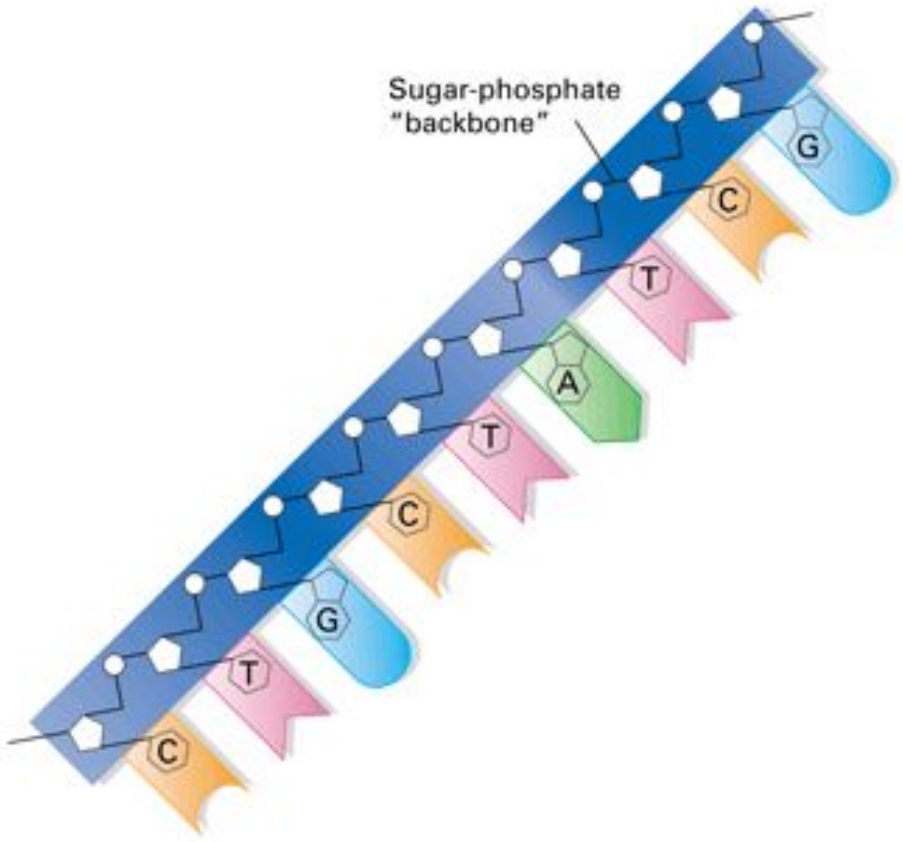
What is the difference between ribose and deoxyribose?

A1.2.3—Sugar–phosphate bonding and the sugar–phosphate “backbone” of DNA and RNA

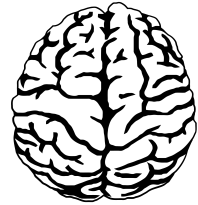
To make the polymer nucleotides are joined together via a covalent bond.

Nucleotides are joined together by adding the phosphate group of a new nucleotide to the 3’ carbon of the previous nucleotide.

This linkage creates a **phosphate-sugar backbone**


Sugar-phosphate
"backbone"
G
C
T
A
T
C
G
T
C

A1.2.4—Bases in each nucleic acid that form the basis of a code

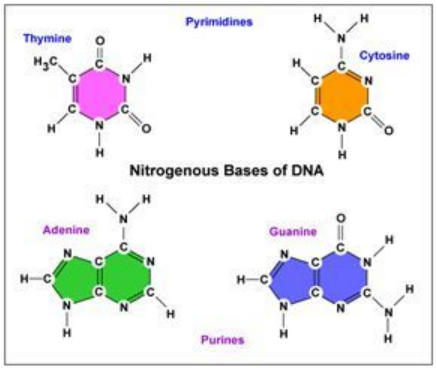


What are the four possible nitrogenous bases in nucleic acid?

4 Nitrogenous Bases

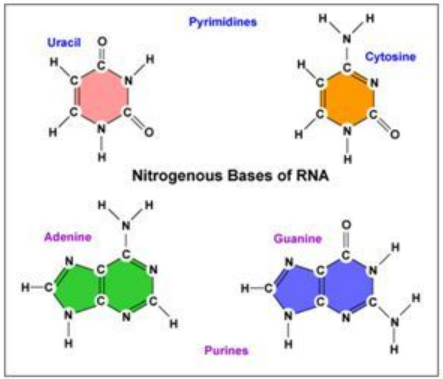
in DNA

- A = adenine
- T = thymine
- C = cytosine
- G = guanine



in RNA

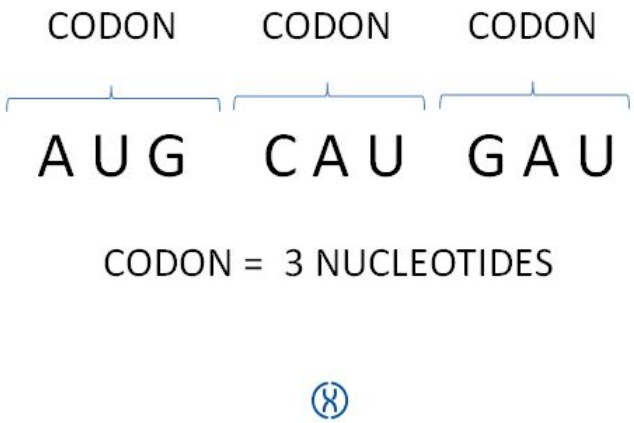
A = adenine
U =uracil (replaces T in DNA)
C= cytosine
G = guanine



A **codon** is made up of a string of three bases, e.g AAT

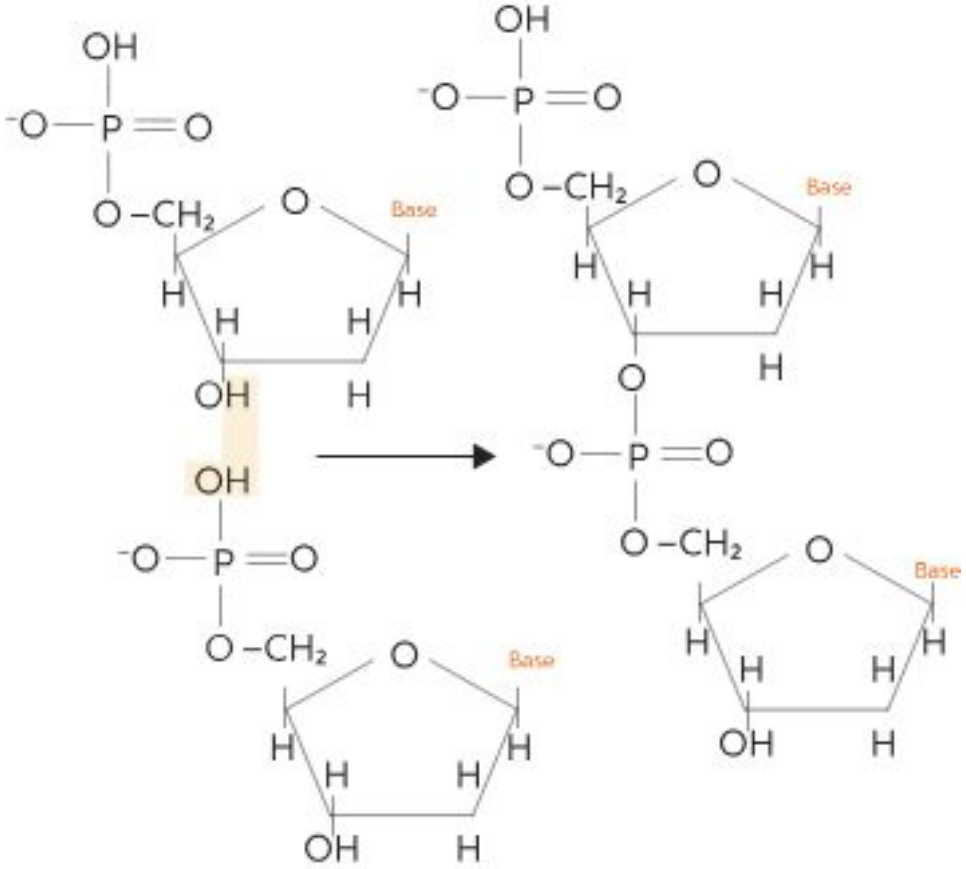
A codon codes for a specific amino acid (or a start or stop codon)

The collection of codons that add up to make a protein is called a **gene**



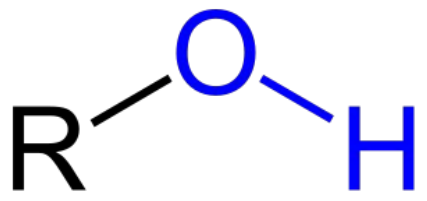
The covalent bond between the phosphate group of one nucleotide and the 3’ carbon of another forms the sugar phosphate backbone

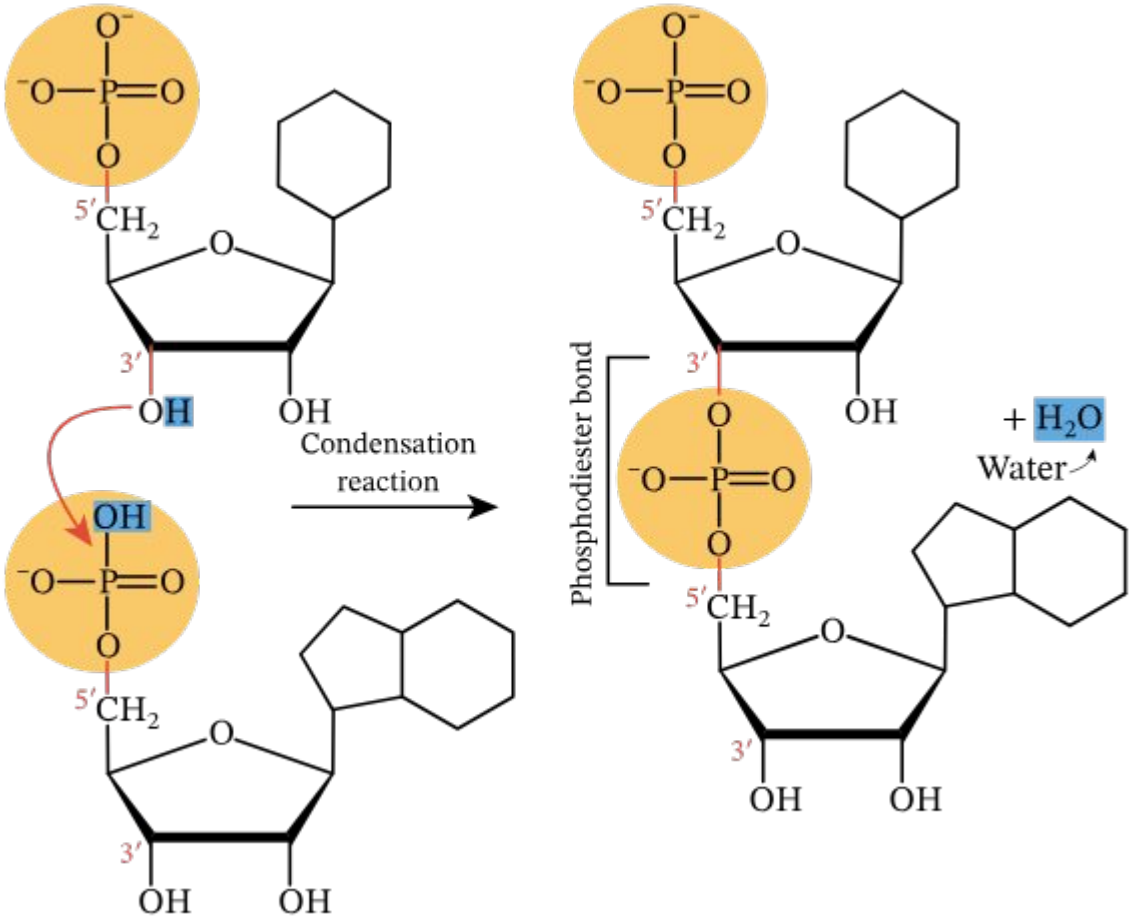
Can you see which atoms are missing when the covalent bond is formed?



A1.2.5—RNA as a polymer formed by condensation of nucleotide monomers

Hydroxyl group





The opposite of a condensation reaction is hydrolysis - more later!



Draw a diagram of a condensation reaction between two nucleotides of deoxyribose.

The base can be represented as a rectangle but you must fully label all the atoms on the deoxyribose and the phosphate

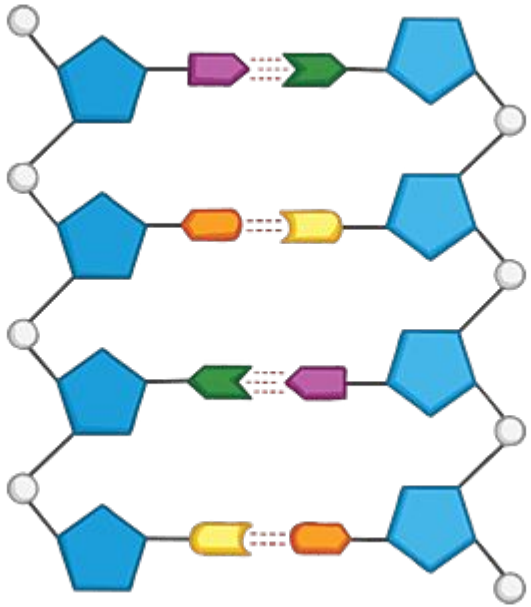
A1.2.6—DNA as a double helix made of two antiparallel strands of nucleotides with two strands linked by hydrogen bonding between complementary base pairs

While RNA can be single or double stranded, DNA is almost always double stranded.

Double stranded

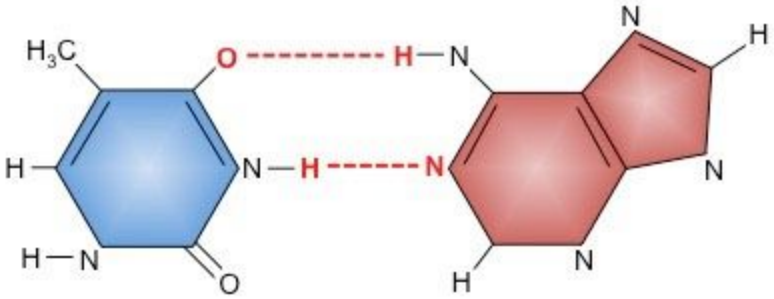
Helix

Antiparallel

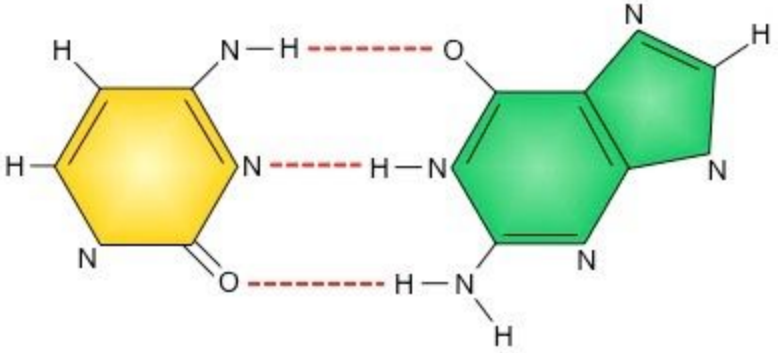


twinkl.com

Complementary base pairing



Thymine / Uracil pairs with Adenine
(2 hydrogen bonds)

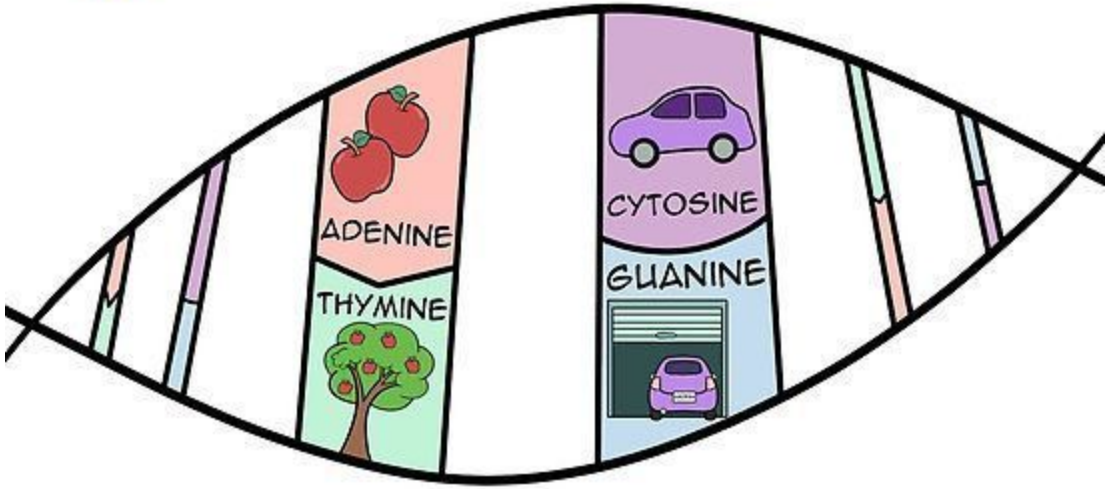


Cytosine pairs with Guanine
(3 hydrogen bonds)

"T for two"

C and G sound like three

"Apples in the Tree"



"Car in the Garage"

The Amoeba Sisters



Using a simple rectangle, pentagon, circle diagram, draw the antiparallel double helix with one side consisting of Guanine, Cytosine and Thymine.

(you may find it easier to draw one side and then turn your paper around to draw the opposite side!)

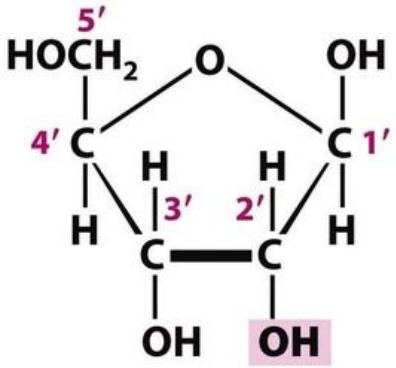
A1.2.7—Differences between DNA and RNA

1. DNA is **usually** double stranded
   RNA is **usually** single stranded
- mRNA
- tRNA
- rRNA

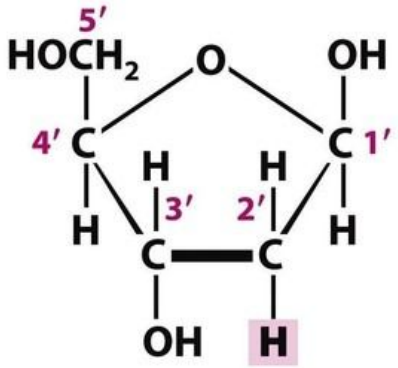
(exceptions include various viruses)

2. DNA always contains **Thymine**
   RNA always contains **Uracil**

3. DNA contains **deoxyribose**
   RNA contains **ribose**



Ribose



Deoxyribose

A1.2.8—Role of complementary base pairing in allowing genetic information to be replicated and expressed

The structure of DNA is ideal for

Replication

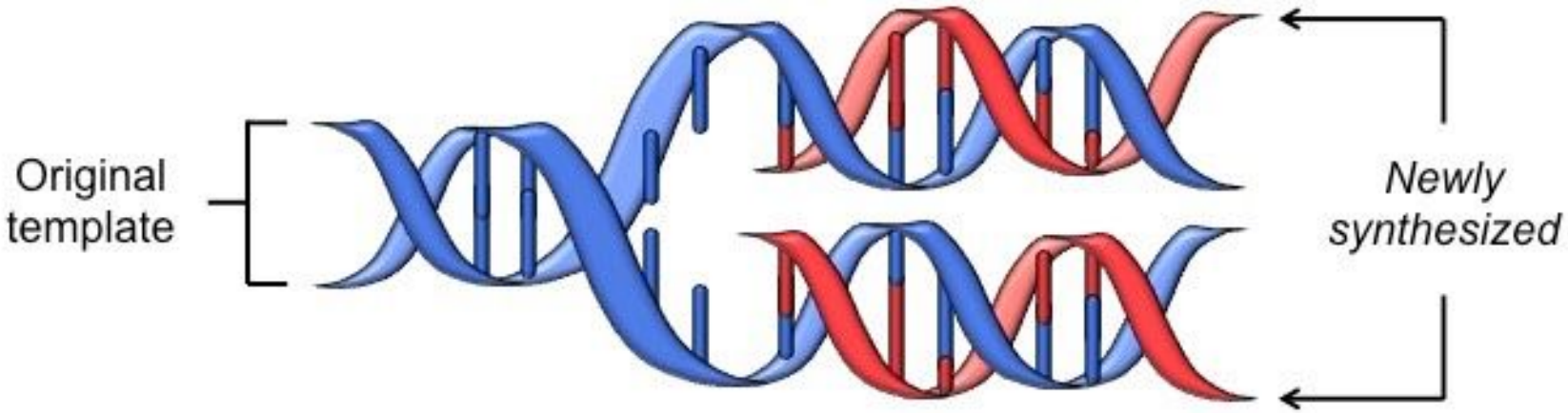
Transcription



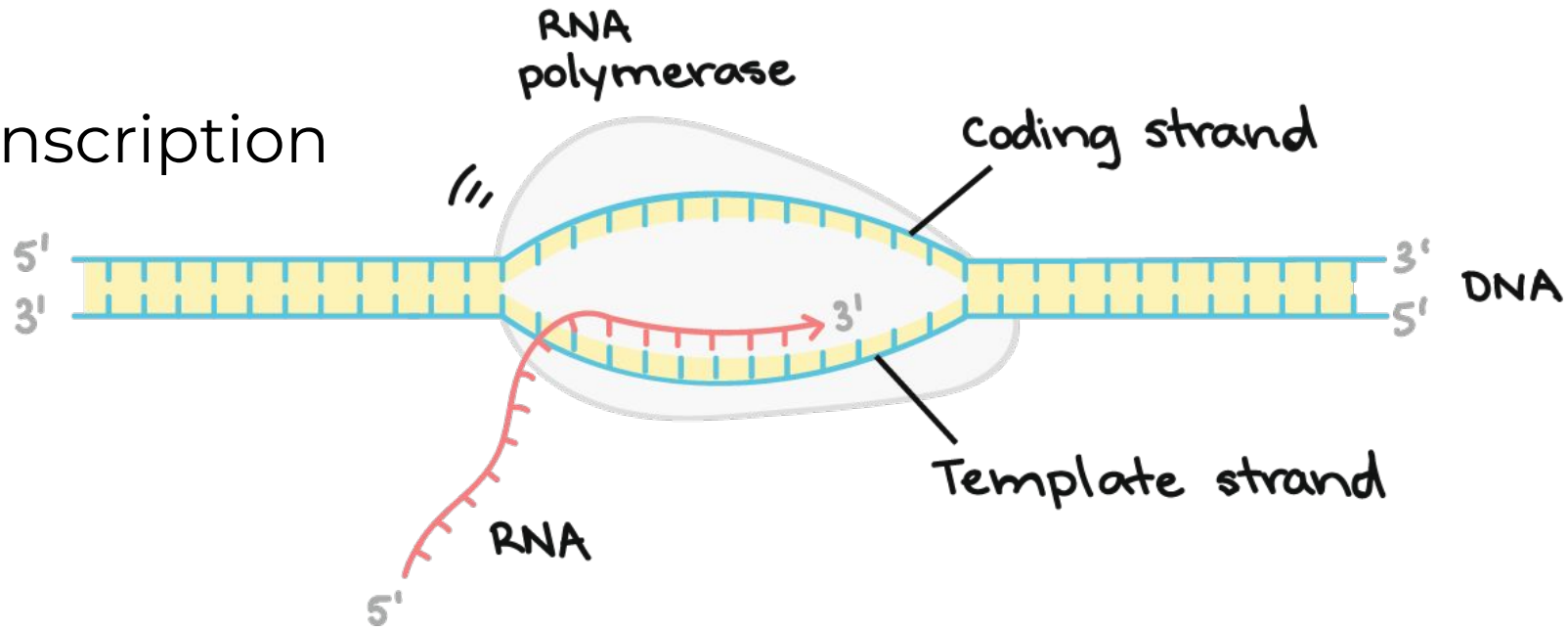
“Genetical Implications of the Structure of Deoxyribonucleic Acid” (1953), by James Watson and Francis Crick

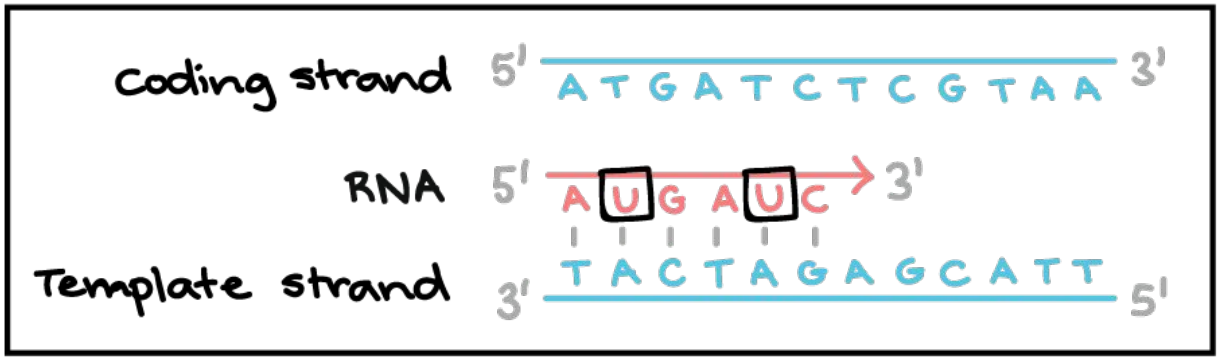
Can you think of a simple way to replicate a double helix of DNA?

Semi conservative replication



Transcription





A1.2.9—Diversity of possible DNA base sequences and the limitless capacity of DNA for storing information

DNA is an excellent molecule for data storage

Incredibly compact (diameter of 2 nanometers)

Limitless storage capacity - the more bases added the more possible combinations.

Just 10 bases give 4^{10} - over a million different combinations!



What does the fact that all organism use exactly the same data storage method tell us about evolution?

A1.2.10—Conservation of the genetic code across all life forms as evidence of universal common ancestry

The genetic code is **universal**

If you place a human gene in a bacteria it will be able to read and express that gene!

Can you...



List the components of a nucleotide?

State the structural difference between ribose and deoxyribose?

List the five possible bases

Explain what is meant by a codon?

Describe how the sugar-phosphate backbone is formed?

Describe the structure of DNA?

Describe complementary base pairing?

Compare and contrast DNA and RNA?

Explain why DNA is such a good data storage medium?

Explain what makes DNA universal?